Symmetric Random Walks with Applications

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Introduction

- Random Walk
- History of Random Walk
  - Karl Pearson
  - Louis Bachelier's *The Theory of Speculation*
  - Albert Einstein
- Recurrence and Transience
  - Polya's Recurrence Theorem.
    Simple random walks of dimension $d = 1, 2$ are recurrent, and of $d \geq 3$ are transient.

Theory

Preliminary

Let

$$a_n \sim b_n$$

denote that

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1$$

Note: When a_n and b_n are both positive sequences:

$$(a_n \sim b_n) \rightarrow$$

$$\left(\sum_n^\infty a_n < \infty \longleftrightarrow \sum_n^\infty b_n < \infty\right) OR$$

$$\left(\sum_n^\infty a_n = \infty \longleftrightarrow \sum_n^\infty b_n = \infty\right)$$

1-Dimensional case

- Markov chain (M.C): $\{X_n | n = 0, 1, ...\}$
- Repulsive barriers: $P_{0,1} = P_{N,N-1} = 1$
- Transition probabilities $P_{i,i+1} = p = 1 - P_{i,i-1} = 1 - q$ for $i = 1, ..., N-1$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & \dots & 0 \\ q & 0 & p & 0 & 0 & \dots & 0 \\ 0 & q & 0 & p & 0 & \dots & 0 \\ \vdots & & & \ddots & & & 0 \\ \vdots & & & & \ddots & & 0 \\ 0 & \dots & \dots & \dots & q & 0 & p \\ 0 & 0 & 0 & 0 & \dots & 1 & 0 \end{bmatrix}$$

1-Dimensional case

It can be shown that

$$P_{rr}^{2n} \sim \frac{[4p(1-p)]^n}{\sqrt{\pi n}}, \quad r = 1, ..., N-1$$

Since $p(1-p) \leq \frac{1}{4}$ for all $p \in \mathbb{R}$, with equality iff $p = 1/2$,

$$\sum_{n=1}^{\infty} P_{rr}^{2n} = \sum_{n=1}^{\infty} P_{rr}^{n} = \infty \quad \text{when } p = 1/2$$

$$\sum_{n=1}^{\infty} P_{rr}^{2n} = \sum_{n=1}^{\infty} P_{rr}^{n} < \infty \quad \text{when } p \neq 1/2$$

Therefore, in 1 dimension:

- A **symmetric** random walk is **recurrent**.
- An **asymmetric** random walk is **transient**.

2-Dimensional case

Consider 2-D symmetric random walk:

$$P_{\langle i,j\rangle,\langle i,j+1\rangle} = P_{\langle i,j\rangle,\langle i,j-1\rangle} = P_{\langle i,j\rangle,\langle i+1,j\rangle} = P_{\langle i,j\rangle,\langle i-1,j\rangle} = 1/4$$

In order for M.C to return to original state in $2n$ steps, must take k steps left, k steps right, $n-k$ steps up, and $n-k$ steps down. The probability this occurs is:

$$\begin{aligned}
P_{\mathbf{rr}}^{2n} &= \sum_{k=0}^{n} \frac{(2n)!}{k!k!(n-k)!(n-k)!} \left(\frac{1}{4}\right)^{2n} \\
&= \sum_{k=0}^{n} \frac{(2n)!n!n!}{n!n!k!k!(n-k)!(n-k)!} \left(\frac{1}{4}\right)^{2n} \\
&= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} \\
&= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n}^{2}
\end{aligned}$$

2-Dimensional case

$$P_{\mathbf{rr}}^{2n} = \left(\frac{1}{4}\right)^{2n} \binom{2n}{n}^2$$
$$\sim \left(\frac{1}{4}\right)^{2n} \left(\frac{4^{2n}}{\pi n}\right) = \frac{1}{\pi n}$$

Since $\frac{1}{\pi}\sum_{n=1}^{\infty}\frac{1}{n} = \infty$, we have that

$$\sum_{n=1}^{\infty} P_{\mathbf{rr}}^{2n} = \sum_{n=1}^{\infty} P_{\mathbf{rr}}^{n} = \infty$$

Therefore, a symmetric random walk is **recurrent** in 2-dimensions.

3-dimensional case

For a symmetric 3-D random walk, we have that

$$P_{\langle i,j,k\rangle,\langle i+1,j,k\rangle} = P_{\langle i,j,k\rangle,\langle i-1,j,k\rangle} =$$
$$P_{\langle i,j,k\rangle,\langle i,j+1,k\rangle} = P_{\langle i,j,k\rangle,\langle i,j-1,k\rangle} =$$
$$P_{\langle i,j,k\rangle,\langle i,j,k+1\rangle} = P_{\langle i,j,k\rangle,\langle i,j,k-1\rangle} = 1/6$$

In order for M.C to return to original state in $2n$ steps, it must take c steps left, c steps right, d steps up, d steps down, $n-c-d$ steps forward, and $n-c-d$ steps back. The probability of this occurring is:

$$P_{\mathbf{rr}}^{2n} = \sum_{\substack{c,d \\ c+d\leq n}} \frac{(2n)!}{c!c!d!d!(n-c-d)!(n-c-d)!}\left(\frac{1}{6}\right)^{2n}$$
$$= \left(\frac{1}{2^{2n}}\right) \sum_{\substack{c,d \\ c+d\leq n}} \left(\frac{1}{3^{2n}}\right) \frac{(2n)!n!n!}{n!n!c!c!d!d!(n-c-d)!(n-c-d)!}$$

3-dimensional case

$$
\begin{aligned}
P_{\mathbf{rr}}^{2n} &= \left(\frac{1}{2^{2n}}\right) \sum_{\substack{c,d \\ c+d\leq n}} \left(\frac{1}{3^{2n}}\right) \frac{(2n)!n!n!}{n!n!c!c!d!d!(n-c-d)!(n-c-d)!} \\
&= \left(\frac{1}{2^{2n}}\right) \binom{2n}{n} \sum_{\substack{c,d \\ c+d\leq n}} \left(\frac{1}{3^n} \frac{n!}{c!d!(n-c-d)!}\right)^2 \\
&\leq \left(\frac{1}{2^{2n}}\right) \binom{2n}{n} \left(\frac{1}{3^n}\right) \frac{n!}{\left[\left(\frac{n}{3}\right)!\right]^3} \sum_{\substack{c,d \\ c+d\leq n}} \left(\frac{1}{3}\right)^n \frac{n!}{c!d!(n-c-d)!} \\
&= \left(\frac{1}{2^{2n}}\right) \binom{2n}{n} \left(\frac{1}{3^n}\right) \frac{n!}{\left[\left(\frac{n}{3}\right)!\right]^3}
\end{aligned}
$$

3-dimensional case

By Stirling's approximation,

$$\frac{n!}{\left[\left(\frac{n}{3}\right)!\right]^3} \sim 3^{3/2}(2\pi n)^{-1}3^n$$

With this, we have that

$$\begin{aligned} P_{\mathbf{rr}}^{2n} &\leq \left(\frac{1}{2^{2n}}\right)\binom{2n}{n}\left(\frac{1}{3^n}\right)\frac{n!}{\left[\left(\frac{n}{3}\right)!\right]^3} \\ &\sim 3^{3/2}(\pi n)^{-1/2}(2\pi n)^{-1} \\ &= \frac{1}{2}\left(\frac{3}{\pi}\right)^{3/2}\frac{1}{n^{3/2}} \end{aligned}$$

3-dimensional case

$$\frac{1}{2}\left(\frac{3}{\pi}\right)^{3/2}\sum_{n=1}^{\infty}\frac{1}{n^{3/2}} < \infty$$

Thus,

$$\sum_{n=1}^{\infty} P_{\mathbf{rr}}^{2n} < \infty$$

Therefore, the symmetric random walk in 3 dimensions is **transient**.

"A drunken sailor will find his way home, but a drunk bird may get lost forever"

- Shizuo Kakutani

Application

Preliminary

Suppose $f(x)$ & $g(x)$ are two functions defined on some subset of $\mathbb{R}$.

Let

$$f(x) = O(g(x))$$

denote that there exists some non-zero constants c and n for which

$$|f(x)| \leq c|g(x)| \text{ for all } x > n$$

Example:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + ... \\ &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + O(x^4) \end{aligned}$$

Diffusion

Suppose a point particle is free to move along a 1-dimensional line. Let $p(x,t)$ represent the probability that this particle is at position x at time t.
For time steps Δt and position steps Δx, let the motion of the particle be modelled by a symmetric random walk:

$$p(x, t+\Delta t) = \frac{1}{2}p(x-\Delta x, t) + \frac{1}{2}p(x+\Delta x, t)$$

Assuming that $p(x,t)$ is a smooth function of both x and t, we may Taylor expand both sides to acquire:

$$p(x,t) + \frac{\partial p}{\partial t}\Delta t + O(\Delta t^2) = p(x,t) + \frac{1}{2}\frac{\partial^2 p}{\partial x^2}(\Delta x^2) + O(\Delta x^4)$$

Diffusion (continued)

Cancelling $p(x,t)$ on both sides...

$$\frac{\partial p}{\partial t}\Delta t + O(\Delta t^2) = \frac{1}{2}\frac{\partial^2 p}{\partial x^2}(\Delta x^2) + O(\Delta x^4)$$

$$\frac{\partial p}{\partial t} + \frac{O(\Delta t^2)}{\Delta t} = \frac{1}{2}\frac{\partial^2 p}{\partial x^2}\left(\frac{\Delta x^2}{\Delta t}\right) + \frac{O(\Delta x^4)}{\Delta t}$$

$$\lim_{\substack{\Delta x \to 0 \\ \Delta t \to 0}}\left[\frac{\partial p}{\partial t} + \frac{O(\Delta t^2)}{\Delta t}\right] = \lim_{\substack{\Delta x \to 0 \\ \Delta t \to 0}}\left[\frac{1}{2}\frac{\partial^2 p}{\partial x^2}\left(\frac{\Delta x^2}{\Delta t}\right) + \frac{O(\Delta x^4)}{\Delta t}\right]$$

$$\frac{\partial p}{\partial t} = \frac{1}{2}\frac{\partial^2 p}{\partial x^2} \tag{1}$$

where we have insisted that $\left(\frac{\Delta x^2}{\Delta t}\right) = 1$ in the limit.

Diffusion (continued)

Let $p(x, t|x_0, t_0)$ be the probability distribution that the particle is at position x at time t, given that it previously inhabited position x_0 at time t_0. Suppose the particle has definite position at time t_0:

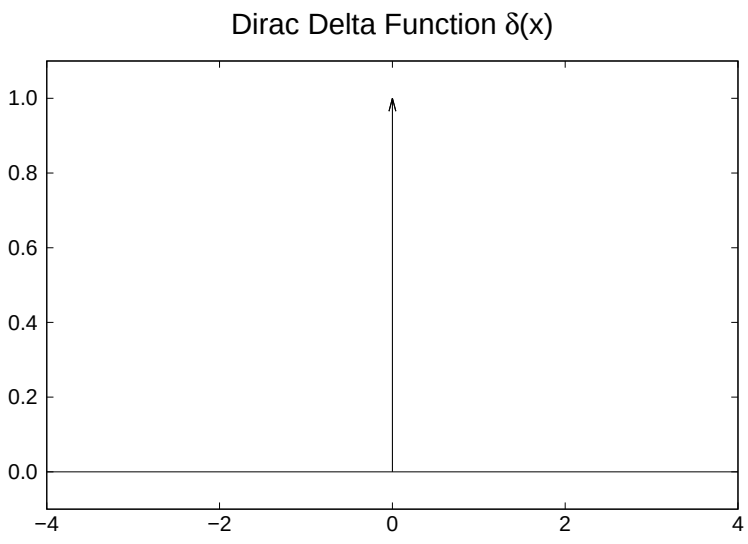
$$p(x, t_0|x_0, t_0) = \delta(x - x_0)$$

where $\delta(.)$ is the Dirac delta function, which may be defined as the derivative of the Heaviside step function $H(.)$:

$$\delta(x) = \frac{d}{dx}[H(x)]$$

where

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$


Dirac Delta Function δ(x)
1.0
0.8
0.6
0.4
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0.0
−4
−2
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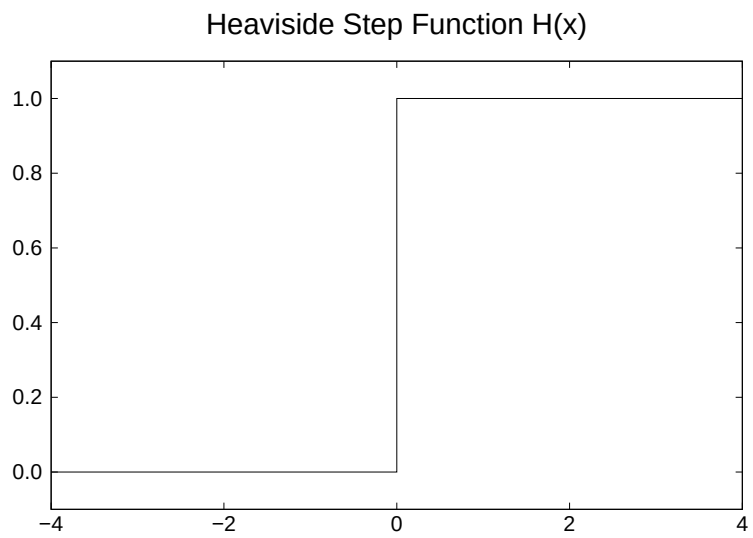
2
4

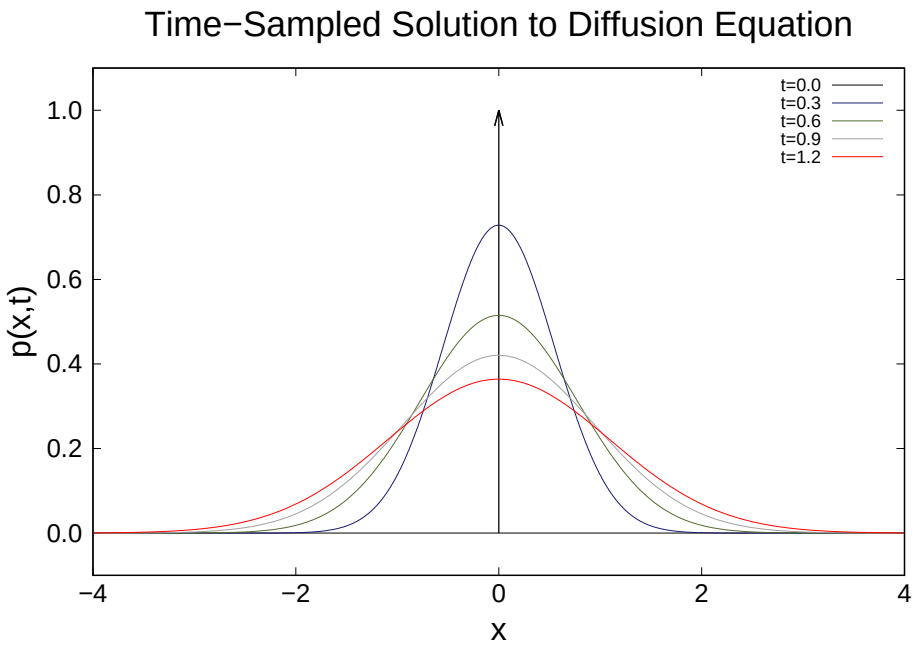

Heaviside Step Function H(x)
1.0
0.8
0.6
0.4
0.2
0.0
−4
−2
0
2
4

Diffusion (continued)

Given this initial condition, equation (1) may be solved using Fourier transforms to obtain:

$$p(x, t|x_0, t_0) = \frac{1}{\sqrt{2\pi(t - t_0)}} e^{\left[\frac{(x-x_0)^2}{2(t-t_0)}\right]}$$

What does this look like?



Diffusion (continued)

Let $P_\epsilon(\Delta t)$ be the probability that the particle escapes a region of size $\epsilon > 0$ in time Δt. We have that:

$$
\begin{aligned}
P_\epsilon(\Delta t) &= \int_{|x-z|>\epsilon} p(x, t+\Delta t|z, t)dx \\
&= \int_{|x-z|>\epsilon} \frac{1}{\sqrt{2\pi\Delta t}} e^{\frac{-(x-z)^2}{2\Delta t}} dx \\
&= 2\int_{\epsilon}^{\infty} \frac{1}{\sqrt{2\pi\Delta t}} e^{-\frac{y^2}{2\Delta t}} dy \\
&= \frac{2}{\sqrt{\pi}} \int_{\frac{\epsilon}{\sqrt{2\Delta t}}}^{\infty} e^{-u^2} du \\
\lim_{\Delta t \to 0} P_\epsilon(\Delta t) &= \frac{2}{\sqrt{\pi}} \lim_{\Delta t \to 0} \int_{\frac{\epsilon}{\sqrt{2\Delta t}}}^{\infty} e^{-u^2}
\end{aligned}
$$

$$
\boxed{\lim_{\Delta t \to 0} P_\epsilon(\Delta t) = 0}
$$

Diffusion (continued)

Let X(t) be the random position of the particle at time t for a sample path. Recall that:

$$p\left[X(t+\Delta t), t+\Delta t | X(t), t\right] = \frac{1}{\sqrt{2\pi\Delta t}} e^{-\frac{[X(t+\Delta t)-X(t)]^2}{2\Delta t}}$$

Now, for some $k, \Delta t \in \mathbb{R}^+$:

$$\begin{aligned} P\left[\left|\frac{X(t+\Delta t)-X(t)}{\Delta t}\right| > k\right] &= P\left[|X(t+\Delta t)-X(t)| > k\Delta t\right] \\ &= \int_{k\Delta t}^{\infty} \frac{1}{\sqrt{2\pi\Delta t}} e^{\frac{-x^2}{2\Delta t}} dx \end{aligned}$$

$$\lim_{\Delta t \to 0} P\left[\left|\frac{X(t+\Delta t)-X(t)}{\Delta t}\right| > k\right] = \lim_{\Delta t \to 0} \int_{k\sqrt{\Delta t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$\boxed{P[|V(t)| > k] = \frac{1}{2}}$$

Conclusion

- Polya's Recurrence Theorem
- Alternative proofs
  - Doyle's Proof: classical theory of electricity, effective resistance and escape probability
  - Montrol (1956) and Finch (2003) probability of returning to the starting points

d	3	4	5	6	7	8
$p(d)$	0.34	0.19	0.13	0.10	0.08	0.07

- Diffusion application
  - Brownian motion is continuous and and non-differentiable everywhere.

References

1. Ross, Sheldon M. *Introduction to probability models*. Academic press, 2014.
2. Kozdron, Michael. Lecture note titled *An Introduction to Random Walks from Pólya to Self-Avoidance*. Duke University, 1998.
3. Keener, James P., and James Sneyd. *Mathematical physiology*. Vol. 1. New York: Springer, 1998.